Colt Melton

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EDUCATION

Kettering University | Flint, MI July 2023 – Present
Bachelor of Science, Computer Science
Ajou University | Suwon, South Korea February 2025 – July 2025
Exchange Program in Software Engineering

TECHNICAL SKILLS

Languages: **Python**, C#, **C++**, Java, JavaScript, **SQL**, C
Algorithms & Data Structures: **Graph Traversal**, Hash Maps, Trees, Heaps, Sorting/Search, Dynamic Programming
Backend & Systems: **REST APIs**, **Kafka**, **Redis**, **SQL/Relational Databases**, SQLite, Data Modeling, Pytorch
Cloud & Infrastructure: **AWS**, **Azure**, Linux, Docker, **VMware**, Networking, **CI/CD**, Bamboo, Robot Framework
Engineering Software: C#/.NET, Python Automation, OPC UA, GUI Development, Git, Testing
Enterprise Software: SAP, ERP audit support, Linux scripting, SQL queries, data aggregation

PROFESSIONAL EXPERIENCE

CSI Intern
October 6, 2026 – Present
Central Garden & Pet

- Support audit-related corporate software work in SAP and enterprise resource planning systems.
- Write scripts and automate Linux workflows to support corporate software operations and audit tasks.
- Develop SQL queries and data aggregation software to organize and summarize information for audit-related work.

Research & Development Engineer / Scientist Intern
October 2025 – October 6, 2026
Endress+Hauser | Ann Arbor, MI

- Built **Python-based monitoring software** integrating **OPC UA**, live spectral visualization, and diagnostic signals to support engineering **validation**, system health analysis, and customer-facing investigations.
- **Automated virtualized software validation** workflows using **VMware, Bamboo, and Robot Framework,** reducing manual test execution by approximately 4 hours per release cycle through repeatable **CI/CD** test orchestration.
- Developed **debugging** and **diagnostic** tools for Raman analyzer systems, helping engineers investigate validation failures, system behavior, customer issues, and root-cause performance problems.
- Integrated **real-time monitoring**, offline data analysis, and predictive inspection workflows to support spectroscopy software development, analyzer validation, and engineering decision-making.

Automation Software and Calibration Engineer
October 2023 – July 2025
Bosch International Engineering | Farmington Hills, MI

- Developed internal **automation software** using proprietary engineering APIs to analyze ASCET-generated embedded control software, enabling engineers to trace parameter dependencies, inspect generated **C-code relationships**, and debug system behavior more efficiently.
- Data analysis from production and prototype braking systems to investigate controller behavior, sensor interactions, calibration parameters, and vehicle dynamics during validation testing.
- Supported **root-cause investigations** for safety-critical chassis systems using **data analysis, diagnostic tooling**, CAN bus traces, instrumentation, and hands-on vehicle testing.
- Built and supported prototype test infrastructure through CAN diagnostics, wiring harness fabrication, connector termination, electrical troubleshooting, and vehicle integration work.

IT Security & Analytics Work-Study
July 2025 - Present
Kettering University | Flint, MI

- Analyzed simulated phishing campaign data to identify security-awareness gaps, user-risk trends, and organizational training priorities.
- Built executive-level reporting for phishing simulation performance, highlighting department-level weaknesses, repeat-risk patterns, and recommended October training focus areas.
- Translated security event data into an actionable awareness strategy using click-rate/report-rate and campaign trend analysis

RESEARCH EXPERIENCE

Thesis: Machine Learning, Neural Network Raman Spectral Decomposition

- Built a **2.9M-parameter PyTorch 1D U-Net** for Raman spectral decomposition, separating baseline and overlapping peak components across **602 real spectra** with reconstruction performance exceeding **$R^2 > 0.95$.**
- Developed an end-to-end scientific computing pipeline using **PyTorch, NumPy, pandas, SciPy, lmfit, and statsmodels** to detect, isolate, fit, validate, and quantify spectral peak components.
- Scaled automated signal analysis across **19,753 validated peak** matches, converting complex multi-component spectra into physically interpretable features for downstream modeling.

• Built spectrum-level **calibration models** for ethane and propane concentration prediction using extracted peak features and regression-based analysis.

PROJECTS

VMware Automation Platform

• Engineered a **C#/.NET infrastructure automation** platform integrating **VMware REST APIs**, **JSON configuration**, and CI/CD validation workflows to provision, configure, and manage repeatable virtualized test environments.

• Automated **VM lifecycle orchestration**, software installation, and **Robot Framework execution** to support **release validation**, debugging, and engineering test reproducibility.

Distributed Telemetry Pipeline

• Built an **event-driven telemetry system** using **Kafka, Redis**, and **SQL** to ingest **simulated system events**, **cache live state**, and persist **historical data records**.

• Implemented **producer/consumer services**, **Redis-backed recent-state lookup**, **SQL-backed event histor**y, and monitoring views for debugging abnormal behavior.

OPC UA Spectroscopy Dashboard

• Developed a real-time industrial data analysis platform using Python and OPC UA, aggregating live spectroscopy measurements and analyzer health metrics to support **operational monitoring** and **predictive diagnostics.**

LEADERSHIP & ACTIVITIES

• **Kagle Leadership Initiative(KLI)** - Mentor role for support and guidance for leadership and academic success amongst early college and high school students.

• **President, Cliff Hangers Rock Climbing Club** - Scheduling, planning, event management, and budget.

CERTIFICATIONS

• **CompTIA IT Fundamentals (ITF+)**